Creatures of the Province: The effect of property tax reform on firms

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"Municipal governments and special purpose municipal institutions such as school boards are creatures of the provincial government."

Ontario Public School Boards' Assn. v. Ontario (Attorney General), 1997

The Ontario property tax reform of 1998 was the most significant change to the property tax system in Ontario's history, and a model for property tax reforms around the world (The Economist, 2021). The reform introduced market value assessments across the province and forced municipalities to lower property tax rates on non-residential property. The reform shifted the burden of the property tax away from non-residential property in an attempt to help businesses, and reduced heterogeneity between Ontario municipalities (Bird et al., 2012). In a notable example prior to the reform, the industrial property tax rate in Toronto was over double that of neighbouring Mississauga. How did businesses in Ontario respond to this reform and did it have any impact on employment? What can we say about the impact of non-residential property taxes on businesses more generally?

This paper will combine data from the annual financial reports of Ontario municipalities with a longitudinal survey of firms to estimate the impact of non-residential property taxes on employment and export status of firm establishments. Geolocating individual establishments will allow for spatial differencing, which pairs establishments across and within municipal boundaries to determine the effect of property taxation in the style of Duranton et al. (2011). Spatial differencing ensures that any location specific time varying trends are controlled for. To control for any remaining unobserved spatial heterogeneity, an instrumental variable is constructed based on provincial government policy following Smart (2012). Unlike previous papers using a spatial differencing methodology, the results will include both manufacturing and non manufacturing establishments.

I find that that the effects of property tax rates on employment are indistinguishable from zero. There is no evidence that this lack of effect is driven by heterogeneity in property class or establishment size,

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and is robust to changes in the spatial differencing distance. Estimates of the effect of property taxes on export status are significant, however there are indications that unobserved spatial heterogeneity is driving the result. Comparisons with the results from previous literature suggests that the non-residential property tax has a minimal effect on establishments so long as property assessments reflect market values and do not include other forms of capital.

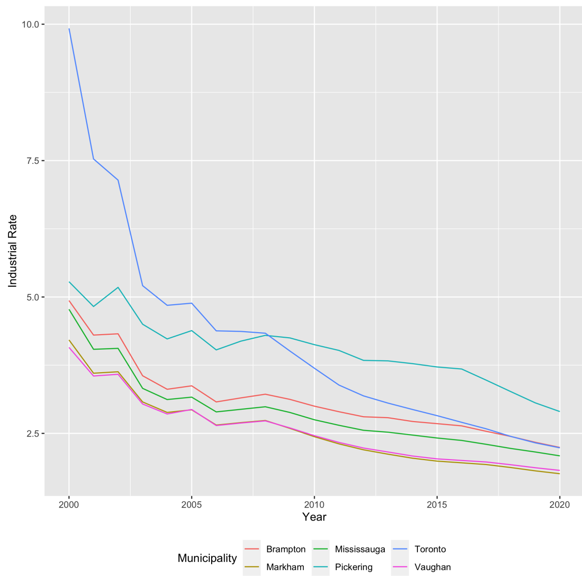
Background

The property tax reform of 1998 was the culmination of a 30 year process to reform the property tax in Ontario. The reform was intended to resolve perceived issues with the property tax system and to ultimately remove the property tax the provincial political agenda. To this end the provincial government undertook three related reforms occurred in the late 1990s: property tax reform in 1998, local services realignment in 1998, and amalgamation of municipalities starting in 1996 and mostly concluding by 2000. The emphasis of this paper is on the tax reform of 1998, while being mindful of the other reforms occurring around the same time.

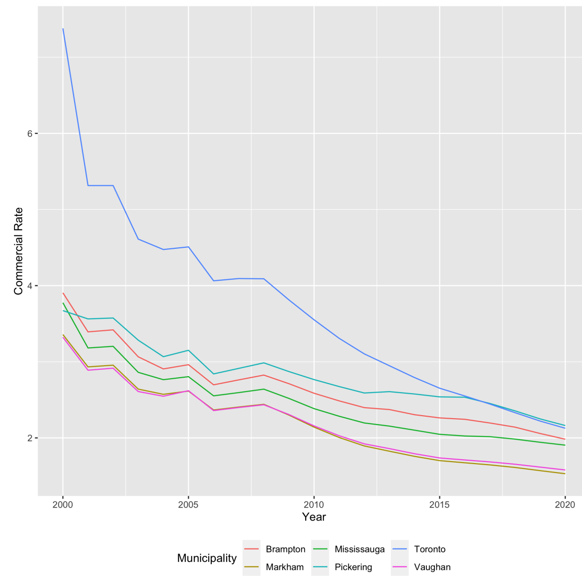
In Ontario the property tax is levied on both residential and non-residential properties by municipalities[1]. Revenue from property taxes is used to fund municipalities and school boards. Municipalities follow a two tier structure, with revenue collected by lower tier municipalities then remitted to upper tier municipalities, or retained in the case of (relatively few) single tier municipalities. Property tax revenue for education is sent to the provincial government, which is then allocated to school boards. The property tax is the largest source of revenue for Ontario municipalities accounting for 47.8% of municipal revenue in 2007 (Bird et al., 2012). Worldwide the property tax amounts to approximately 3% of GDP across the OECD (Slack, 2011).

Prior to 1998, one of the ongoing problems with the property tax was property assessments, which were non uniform across municipalities. The 1998 reform created the Ontario Property Assessment Corporation (later renamed the Municipal Property Assessment Corporation or MPAC) which took over property assessment for the entire province. Assessments were then updated to be based on the Current Value Assessment (CVA) of a property, which should reflect the value if sold to an independent party. Prior to the reform residential properties were undervalued relative to their market value, while commercial properties were valued at approximately their market value (Smart, 2012). Imposing a uniform property tax with CVA assessments would have results in a large shift in the burden of taxations onto residential property. To avoid this shift and the potential fallout from voters, the province established multiple property classes, which municipalities

[1] Bird et al., 2012 is the most comprehensive review of the Ontario property tax reform. Any errors in the background are mine and mine alone.



(a) Industrial property tax rates in the GTA



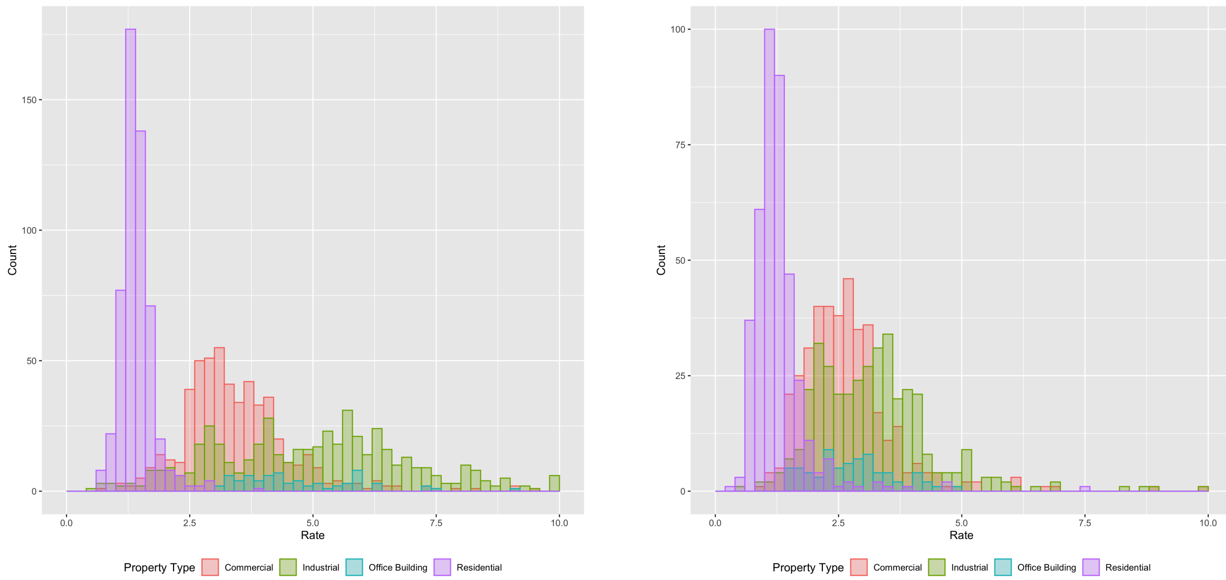
(b) Commercial property tax rates in the GTA

Figure 1: Property tax rates in the GTA after the 1998 property tax reforms.

could tax at different rates. This allowed municipalities to maintain the tax burden on each property class after the assessments were updated, and created a classified system for property taxes.

The province then established rules to regulate the ratio between the residential and non-residential property classes to limit the ability of municipalities to shift the property tax onto businesses. The property tax ratio was defined to the be ratio of the property tax rate for a given class relative to the residential property tax rate. For example, if the property tax rate for residential property was 1.25% and the property tax rate for commercial property was 2.5%, then the property tax ratio would be 2.5%/1.25% = 2.0. The province then established a target range for property tax ratios of between 0.6 - 1.1, depending on the property class. These targets were known as the "range of fairness". Starting in 1998, municipalities could set property tax rates for each class that moved towards the range of fairness, but not rates that moved away from it. This was intended to limit municipalities ability to shift the property tax burden onto businesses.

In addition to the range of fairness and in response to complaints from the business community, the province imposed further caps on property tax increases for the commercial, industrial, and multi-residential classes in 2001. The province defined threshold tax ratios for three property classes: 1.98 for commercial, 2.63 for industrial, and 2.74 for multi-residential. Municipalities above these cutoffs could only increase their tax rates in that property class after their tax ratio fell below those thresholds (Smart, 2012). In 2004 the province relaxed this restriction, allowing municipalities to increase tax rates for the the commercial, industrial, and multi-residential classes only to the extent that it preserved the tax share of that class for the municipality. This option is known as "tax ratio flexibility", combined with the threshold ratios for the



(a) 2000 (b) 2020

Figure 2: Property tax rates across Ontario, by type, between 2000 and 2020.

commercial, industrial, and multi-residential property classes, remains in place as of the time of writing[2].

The policy of threshold ratios formed a "Hard Cap" policy (Smart, 2012) that forced municipalities to reduce their commercial and industrial tax rates if they were above the threshold. Figure 1 shows ndustrial and commercial property tax rates across the Greater Toronto Area (GTA) starting in 2000. There is a notable drop in the rates in Toronto relative to the rest of the GTA as Toronto experienced this hardcap policy. Across all of Ontario there was a clear narrowing of the distribution of commercial and industrial rates. Figure 2 shows the distribution of property tax rates across Ontario between 2000 and 2020. The narrowing of the distribution is largest and most notable in the industrial property class.

Simultaneously the provincial government engaged in local services realignment which shifted spending responsibilities between the province and municipalities. Many services such as policing, airports, and sewer and water which were previously joint between the province and municipalities were downloaded to be solely municipal responsibilities, with a smaller number of services being uploaded to the province. In exchange the province took over education financing, which was previously handled by local school boards. Instead of municipalities setting education tax rates for their local school board, the province would set education tax rates, collect the revenue, then redistribute it to the school boards as needed. After local services realignment the province set a uniform education tax rate across all residential properties in province. The province retained existing differences in the education property tax burden on non-residential properties, and set the Business Education Tax (BET) accordingly. Over time the province has attempted to decrease

[2]I am grateful to Katherine Chu and Aidan Carter at the Ministry of Finance for their explanation of this system

the differences in BET rates across municipalities (Bird et al., 2012).

A further confounding change to the effect of the property tax reform was the amalgamation of municipalities. In 1996 Ontario passed the Savings and Restructuring Act which was designed to encourage municipalities to merge to make local governments more efficient. From 1996 to 2004 the number of municipalities in Ontario reduced from 815 to 445. The majority of these amalgamations were voluntary, with the forced amalgamation of Toronto in 1998 being a notable exception. The majority of these amalgamation happened prior to 2000, from 2000 to 2004 the number of lower tier and single tier municipalities decreased from only 536 to 415[3].

Literature

The Ontario property tax reforms have been documented in Bird et al. (2012), which covers the history of the Ontario property tax reforms and it's effects on municipalities. Smart (2012) utilizes the reform to estimate the effect of increased business taxation on business location and employment, finding an elasticity of employment with respect to tax rate of -0.23. Smart (2012) concludes that the reform has had a small positive impact on employment and productivity, although is limited by using data aggregated at the municipality level.

This paper is most closely related to the literature on the effect of property taxation on firms starting with Duranton et al. (2011). Duranton et al. (2011) uses microgeographic data from the Annual Census of Production in the UK to estimate the effect of local property tax changes on manufacturing establishments. They find that non-residential property taxes have a negative effect on establishment employment but no effect on firm entry. They estimate an elasticity of employment with respect to the property tax rate of -1.024, suggesting that property taxes can have significant effects on employment. A key caveat is an institutional features of the UK, where property reassessment occur if buildings are expanded. Therefore the effect of property tax rate changes on employment are larger as firms fail to expand to avoid reassessments on existing properties. Comparing to Ontario, where the assessed value of properties are regularly updated after the 1998 reform, this mechanism will not be relevant.

Belotti et al. (2021) expands on the analysis of Duranton et al. (2011), finding that the Business Property Tax (BPT) in Italy has a negative impact on equipment, employment and value added on Italian manufacturing firms. BPT rates include both the value of land and buildings as well as equipment and machinery in the assessment tax base, and they find that there is no measurable change in the usage of land and buildings as a result of BPT changes. They go on to show that changes in employment due to BPT rates

[3]Based on Financial Information Return from the Government of Ontario

are attributable to changes in equipment and machinery. A limiting factor of Belotti et al. (2021) is that multi-establishment firms are excluded.

Both Duranton et al. (2011) and Belotti et al. (2021) make use of a technique called spatial differencing, which compares pairs of firms (dyads) across and within municipal boundaries. Cameron and Miller (2014) examines how to perform robust inference on dyadic data. Belotti et al. (2018) examines robust inference in spatial differencing models specifically, and shows how the dyadic robust estimator used in Cameron and Miller (2014) supersedes that of Duranton et al. (2011).

A related literature explores the effect of tax differentials in the presence of agglomeration externalities. Brülhart et al. (2012) utilize data from Swiss municipalities to show that agglomeration forces offset the effect of tax differentials on firm entry. For a review of the literature, see Brülhart et al. (2015). This paper will also make use of the work of Behrens and Bougna (2015) to document the geographical concentration of manufacturing in Canada.

This paper also relates to the literature on firm location choices under taxation. Jofre-Monseny and Solé-Ollé (2010) examines the effect of local taxation on new manufacturing firms, finding small effects from non-residential property taxes relative to local business taxes. Giroud and Rauh (2019) estimates the effect of state taxation on business activity, finding that the corporate taxes reduce employment and the number of establishments for C corporations, although pass-through entities are less sensitive with respect to the comparable personal tax rate. Fajgelbaum et al. (2019) uses state tax rate changes to show that firms are sensitive to after tax expected profits, and utilize this result to calibrate a model with facing different taxation mixes. They find that the heterogeneity is state taxes is a significant source of welfare loss in the United States.

Empirical Strategy

The empirical strategy of this paper will mirror that of Duranton et al. (2011), making use of the improved matching structure developed in Belotti et al. (2021). To motivate the model, consider a regression of the property tax rate on some outcome of interest for establishments:

$$e_{it} = \alpha r_{at} + X_{it}\beta + \mu_i + \epsilon_{it} \tag{1}$$

Where e_{it} is the outcome of interest (employment or export status) of establishment i at time t, r_{at} is the property tax rate of municipality a at time t, and α is the coefficient of interest. X_{it} is a vector of firm controls, and μ_i is some unobservable establishment characteristic which is invariant over time. This model is likely to be biased as the locations of establishments are likely to be heterogenous. Establishments could vary due to municipality specific factors, like tax or spending policies, which can be thought of an unobservable municipality characteristics γ_a. More importantly there are likely to be location characteristics which occur at a much finer geographic scale, such as proximity to roads, and which vary continuously over space and time. These features are modelled as a parameter, θ_{zt}. The model then becomes:

$$e_{it} = \alpha r_{at} + X_{it}\beta + \mu_i + \gamma_a + \theta_{zt} + \epsilon_{it}. \tag{2}$$

Estimating equation 2 via OLS is likely to return biased results due to the unobserved factors μ_i, γ_a, and θ_{zt}. Duranton et al. (2011) resolve this issue by making use of two transformations to account for unobserved factors. First take the within transformation on equation 2 to remove any time invariant features. This gives:

$$\tilde{e}_{it} = \alpha\tilde{r}_{at} + \tilde{X}_{it}\beta + \tilde{\theta}_{zt} + \tilde{\epsilon}_{it}, \tag{3}$$

where variables with a $\sim$ have had the within transformation applied.

The resulting equation still contains $\tilde{\theta}_{zt}$, for time varying spatial unobserved heterogeneity. To control for this Duranton et al. (2011) performs spatial differencing between pairs of establishments located close to each other. Spatial differencing matches firms with similar characteristics both within and across municipal boundaries, within some distance. Duranton et al. (2011) considers establishments to be similar if they operate within the same 2 digit industry, while Belotti et al. (2021) considers firms as being similar if they are in the same 2 digit industry and in the same production quintile. The distance within which establishments are matched is chosen by the researcher and in practice is a tradeoff between the number of matches (larger distances create more matches) versus the ability to control for spatial heterogeneity $\tilde{\theta}_{zt}$. If firms are located closely together then the difference in any spatially continuous unobserved factor between the two firms should be approximately zero. Performing spatial differencing recovers:

$$\Delta_d\tilde{e}_{jt} = \alpha\Delta_d\tilde{r}_{at} + \Delta_d\tilde{X}_{jt}\beta + \Delta_d\tilde{\theta}_{zt} + \tilde{\epsilon}_{jt}. \tag{4}$$

where Δ_d is the spatial differencing operator. Assuming spatial differencing is performed over small enough distances the term $\Delta_d\tilde{\theta}_{zt} = 0$, resulting in:

$$\Delta_d\tilde{e}_{jt} = \alpha\Delta_d\tilde{r}_{at} + \Delta_d\tilde{X}_{jt}\beta + \tilde{\epsilon}_{jt}. \tag{5}$$

After spatially differencing the unit of observation changes to be by establishment pair j, instead of by establishment i. This creates a dyadic data structure, necessitating a correction of the standard errors to account for the correlation between establishment pairs which contain the same establishment. Belotti et al. (2018) discusses the necessary correction to standard error to obtain robust inference.

As demonstrated in Duranton et al. (2011) spatial differencing may not fully control for spatial features in practice as matching may occur at such a large distance that shocks to local conditions may still be correlated within municipalities. As a final stage I therefore instrument the transformed tax rate $\Delta_d \tilde{r}_{at}$ to recover unbiased estimates of α. Given the pairwise structure of the data from spatial differencing and the within transformation having been performed, a successful instrument must therefore be varying over both space (between municipalities) and time. I therefore make use of the Hard Cap policy in Ontario, which affected some municipalities in each year. The instrument is then transformed in same manner (within transformation then spatially differenced) as the other variables.

Data

To implement the estimation strategy requires detailed data on both the property tax in Ontario, as well as panel data of establishments and establishment outcomes. Data on the property tax by municipality can be obtained from the Financial Information Return (FIR) from the provincial Ministry of Municipal Affairs and Housing (Government of Ontario, 2022). Each municipality is required to submit the FIR each year to the ministry. FIRs are available from 1988 to the present, although only FIRs from 2000 onward are practical due to the large number of municipal amalgamations in the years immediately after 1996. Panel data on firms comes from Scott's National All Business Directories Database, a directory of businesses in Canada[4]. The Scott's database is utilized in Behrens and Bougna (2015) to compute geographic concentration indices by industry in Canada. The data cover 2001 to 2019 in two year increments, with 2015 being absent. Panel identifiers are only available from 2003 onwards, so 2003 - 2019 is the sample period.

From the FIRs I assemble a panel tax data containing the aggregate assessed value, tax rate, paid taxes, and the tax ratio for each property class in each municipality. In cases where there are multiple property subclasses, for example for unoccupied buildings, payments in lieu[5], or separate rates reflecting recent municipal amalgamations, subclasses are aggregated together. I focus on the industrial and commercial property classes, as these were the classes to which the Hard Cap policy applied. All tax variables include both upper tier municipality and lower tier municipality rates, aggregated into a single municipal value. Education tax rates and paid taxes are split out separately. Summary statistics for tax variables can be found in table 12 in the appendix.

[4]I would like to thank Kristian Behrens for graciously sharing this data with me.

[5]Provincial and Federal government buildings are exempt from property taxes, these governments make Payments in Lieu of property taxes (Bird et al., 2012).

With the tax ratio for each property class and municipality, I construct an indicator variable for whether the Hard Cap policy applied in a given year. Given the threshold ratio for the commercial and industrial classes described in the Background section. I can determine whether the Hard Cap policy applied to a given municipality in a given year. An indicator variable for the Hard Cap is created which equals 1 if the Hard Cap policy applied to that municipality in that year.

Population data come from the Statistics Canada annual population estimates by census subdivision Government of Canada (2017). Population data cover 2001 - 2021, and are broken out by census subdivision which correspond approximately to each municipality. There is not an exact matching between the annual population estimates and the FIRs due to municipal amalgamations changing municipal boundaries faster than the census subdivision boundaries. Annual population estimates are matched to the FIRs via a combination of exact matching, regex matching, and fuzzy matching. The overall match rate is 99.3%, the details from the matching procedure can be found in section 0.3 in the appendix. The result of this procedure is both population counts by municipality and links between the municipality identifier in the FIRs and the census subdivision identifier, which will be used to help locate firms within municipalities.

From the FIRs, I also assemble data on municipal expenditures. Unlike taxation, which is administered solely by lower tier municipalities, expenditures are carried out by both the upper and lower tier municipalities. Some expenditure items are the responsibility of the upper tier municipality while other responsibilities are split between tiers, with no consistent allocations of responsibilities between tiers across the province. To estimate municipal expenditures within each lower tier municipality I sum between lower tier expenditures and upper tier expenditures, distributing the upper tier expenditures to the lower tier by population. For upper tier municipalities with both urban and rural lower tier municipalities, this is likely to overstate expenditure in urban areas and understate expenditure in rural areas, where the cost of public service provision is likely higher. Summary statistics for expenditures can be found in table 13 in the appendix.

Data on individual firm establishments come from the Scott's database. To implement the estimation strategy requires firm establishment data with a panel dimension and precise geolocations. The Scott's database contains panel identifiers for individual establishments from 2003 - 2019, including exact addresses and postal codes which can be geolocated. The Scott's database is primarily focussed on the manufacturing industry, with Behrens and Bougna (2015) finding that the database is representative of the manufacturing industry when compared to Canadian Business Patterns (Government of Canada, 2022). Unlike Duranton et al. (2011) and Belotti et al. (2018) I include firms in both manufacturing and non-manufacturing, as determined by the NAICS code. A drawback of the Scott's database is that it is based on surveys, requiring more extensive data cleaning. An advantage is that the database is at the establishment level, ensuring multi-location firms are not a concern as in Belotti et al. (2018).

Table 1: NAICS codes and their corresponding property classes

NAICS Codes	NAICS Industry	Property Class
31 - 33	Manufacturing	Industrial
41 44, 45	Wholesale and Retail Trade	Commercial

Table 2: Sequential results from data cleaning. The number beside each step shows the number of observations remaining after that step of data cleaning is performed.

Step	Number
Raw Observations	609535
Remove 2001	556992
Industrial or Commercial	247344
>2 Observations	237500
Remove moved or municipality changed	148917
Remove property class changed	135326
Missing Municipality or Coordinates	135055
Missing NAICS, Employment, Exporter, Business Type, HO	127553
Employment or Square Footage zero	127544
Employment < 7000	127521
Square Feet per Employee < 75 or > 10ˆ5	127521

I geolocate each establishment in the Scott's database using the Postal Code Conversion File (PCCF) from Canada Post. Following Behrens and Bougna (2015) establishments are geolocated using the PCCF from the following year to account for the six month delay in updating postal codes. In some cases postal codes do not uniquely assign a census subdivision, in those cases string matching is performed between the city listed in the Scott's database and the census subdivision name. In total 99.8% of establishments are geolocated. Matching is then completed between each establishment and and the municipality from the FIRs. 99.5% of establishments are located to a municipality. Establishments are matched to the relevant property class by year, municipality, and NAICS code, as seen in table 1. Details on the matching procedure can be found in section 0.3 in the appendix.

For each establishment I observe the address, NAICS codes, estimates of employment, square footage, export status, and establishment type. The establishment type is a basic description of the business activity at that location (e.g. Manufacturer, Retail, Sales Office). Some variables also have a large number of missing values. Where possible I impute missing values based on previous years. Square footage is missing in nearly half of the observations, as a result I bin square footage into quartiles and include a separate category for firms with "NA" to ensure those observations are not lost.

Several significant restrictions are made to the Scott's database to account for firms which have moved or changed industries. The estimation strategy only applies to firms which remain in the same location after property tax rate changes have occurred. As a result all firms which change location are removed from the sample. In addition it is possible that firms change industries, which could change how firms are matched to property classes as given by table 1. As a result all firms whose property class changes are dropped. In addition to accounting for moving firms, some observations are removed due to other missing variables. The consequence of each set of restrictions can be found in table 2. In total there is a final set of 127,521 observations in the 2003 - 2019 period, prior to any matching occurring.

Results

To show the effects of each transformation, I estimate equations 2, 3, 5, and equation 5 with instrumenting, taking as the outcome variable the establishment employment then the export status of the establishment. The explanatory variable of interest is the municipal property tax rate. As discussed in Background section, the education property tax rate was changed exogenously by the province over this time period, and is highly correlated with the municipal property tax rate. As a result I include log educational property tax rate as a control variable, but this is not the variable of interest.

As firm level controls I include the controls in Duranton et al. (2011) of a second order polynomial of establishment age, and a dummy variable for if the establishment is greater than 20 years old or if the age of the establishment is unknown. A significant number of observations are missing the establishment age due to the Scott's database being a survey, or quote the firm age rather than the establishment age. Taking a 20 year cutoff ensures that errors due to reporting are minimized. I also include an indicator for whether the establishment is the headquarters of the firm, which is time varying in the data.

Three variables for time invariant establishment characteristics are included: 2 digit NAICS industry codes, establishment types, and the quartile of establishment square footage. Time invariant establishment characteristics are included as fixed effects in the OLS and within transformed specifications in equations 2 and 3, and are the variables used to match pairs of firms when performing spatial differencing to estimate equation 5. As a result matching occurs at the industry/establishment type/square footage quartile level.

Taking the within transformation and performing spatial differencing limits the set of establishments which can be included in the estimation. To ensure comparability between estimation strategies, I limit the set of establishments used in estimating equation 2 and 3 to be the same as those used in estimating equation 5. First taking the within transformation removes any establishment with less than two observations. Spatial differencing produces pairs of firms, which leads to any observation which has no corresponding partner to be dropped. Estimating equation 5 will also include a pair fixed effect, requiring that each pair of observations be observed at least twice in the data. As a result the standardized set of establishments are those with at least two observations, matched to at least one other firm with two observations in the same years.

Table 3: Estimations of the elasticity of establishment employment with respect to the municipal property tax rate for 2003 - 2011. Controls include the education property tax rate, age of the establishment, age^2 of the establishment, a dummy if the establishment is greater than 20 years old, and whether the establishment is the head office of the firm. Column (1) reports the estimation using pooled OLS, column (2) takes the within transformation to control for establishment fixed effects, and column (3) performs spatial differencing using a 1km threshold and controls for establishment fixed effects. Column (4) reports the first stage, instrumenting for the municipal tax rate using the transformed Hard Cap policy variable, and column (5) reports the full instrumental variable estimation. Standard errors are clustered at the establishment level in columns (1) and (2), and by both establishments in (3), (4), and (5) as an approximation to optimal dyadic robust standard errors. The relevant coefficients are the municipal rate coefficients in the top row.

Dependent Variables:	Employment (ln)			Mun. Rate (ln)	Employment (ln)
IV stages				First	Second
Model:	(1)	(2)	(3)	(4)	(5)
Variables					
Mun. Rate (ln)	-0.1233***	-0.0002	0.0383		0.0004
	(0.0334)	(0.0218)	(0.2084)		(0.2071)
Educ. Rate (ln)	0.4145***	-0.0308	-0.2480	2.119***	-0.1663
	(0.0885)	(0.0450)	(0.6158)	(0.0392)	(0.5967)
Establishment Age	0.0262*	0.0232***	0.0367***	-6.16×10^{-5}	0.0367***
	(0.0141)	(0.0061)	(0.0080)	(4.37×10^{-5})	(0.0080)
Establishment Age^2	-0.0002	-0.0005**	-0.0008***	2.67×10^{-6}	-0.0008***
	(0.0005)	(0.0002)	(0.0003)	(1.68×10^{-6})	(0.0003)
Old Location Dummy	0.6558***	0.3008***	0.4071***	-0.0003	0.4071***
	(0.0875)	(0.0472)	(0.0632)	(0.0004)	(0.0632)
Headquarter Office	0.6867***	0.0179*	0.0342**	5.78×10^{-5}	0.0342**
	(0.0285)	(0.0105)	(0.0140)	(6.45×10^{-5})	(0.0140)
Hard Cap Policy				0.1414***	
				(0.0041)	
Fixed-effects					
Year	Yes	Yes	Yes	Yes	Yes
2 Digit NAICS	Yes	Yes			
Establishment Type	Yes	Yes			
Square Footage Quartile	Yes	Yes			
Establishment ID		Yes			
Pair			Yes	Yes	Yes
Fit statistics					
F-test (1st stage)				256,108.1	
F-test (1st stage), Mun. Rate (ln)					256,108.1
Observations	50,556	50,556	91,282	91,282	91,282
Establishments	13,948	13,948	13,948	13,948	13,948
Pairs	-	-	29453	29453	29453

*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Table 3 shows the results of equations 2, 3, 5, and equation 5 with instrumenting using the establishment employment as the outcome variable. Column (1) estimates equation 2, using a pooled OLS estimation, with fixed effects for year and each of the time invariant establishment characteristics. The coefficient for the employment on municipal property tax rate is the coefficient of interest and can be interpreted as an elasticity of employment with respect to property tax rate. I find that this elasticity is negative and significant when estimated with pooled OLS. Making use of the panel dimension of the data I then estimate the within specification in equation 3, which is shown in column (2). The within transformation removes any firm fixed effects, after which the result is indistinguishable from zero.

Spatial differencing is then performed and estimated as in equation 5. Spatial differencing is performed using a radius of 1km, analogous to Duranton et al. (2011) and smaller than the 2km used in Belotti et al. (2021), with matching occurring for each of the time invariant establishment characteristics. From the original $13,948$ establishments this produces $29,453$ pairs over time, for a total of $91,282$ observations. The results of the estimation can be found in column (3) of table 3. The estimate of the elasticity of employment with respect to municipal tax rate remains indistinguishable from zero. Reassuringly there are no significant changes in the estimated coefficients for any of the control variables.

Finally I instrument the municipal tax rate with the within transformed, spatially differenced indicator variable for the Hard Cap policy. Column (4) of table 3 shows the first stage of the instrumental variable. The F-stat shows no issue of weak instruments, and the sign for the Hard Cap policy variable is positive, indicating that the policy was applied to places with a higher municipal property tax rate. Column (5) shows the results of the second stage of the estimation. The estimated elasticity of municipal tax rate on employment is found to be essentially zero. The estimated coefficient for the municipal tax rate after instrumenting in column (5) is lower than before instrumenting in column (3), consistent with the instrument accounting for some correlation between local business conditions and property tax rates proposed by Duranton et al. (2011) which was not fully controlled for by spatial differencing. This biases the within transformed and spatial differenced estimate in column (3) upward relative to the instrumented estimation in column (5).

It is possible that the results in table 3 could mask some underlying heterogeneity in responses by establishments.Table 4 provide decompositions by property class and square footage quartile. Columns (2) and (3) of table 4 split the sample into both commercial and industrial property classes, where there are no significant differences between the estimates of the elasticity of employment with respect to the municipal tax rate. Columns (4) through (8) break down firms by their quartile of square footage, and also find no estimates different from zero. These results suggest that heterogeneity in establishments is not masking some underlying variation.

Table 4: Estimations of the elasticity of establishment employment with respect to the municipal property tax rate for 2003 - 2011. Column (1) repeats the estimation in column (4) of table 3. Columns (2) and (3) repeat the same analysis for commercial and industrial property independently. Columns (4) - (8) repeat the analysis for quartiles of square footage, and for establishments missing square footage in the data. Standard errors are clustered by both establishments as an approximation to dyadic robust standard errors. The relevant coefficients are the municipal rate coefficients in the top row.

Dependent Variable:	Employment (ln)							
Property Class	Full sample	Commercial	Industrial					
Square Footage Quartile				1	2	3	4	NA
Model:	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Variables								
Mun. Rate (ln)	0.0004	0.3304	-0.1155	0.0007	-0.1289	-0.3792	-0.0126	0.1257
	(0.2071)	(0.4664)	(0.2296)	(0.1421)	(0.4755)	(0.6050)	(0.4299)	(0.3128)
Educ. Rate (ln)	-0.1663	-0.6953	0.0476	3.555*	-0.0631	-0.1330	0.4491	-0.6142
	(0.5967)	(2.782)	(0.6317)	(1.922)	(1.196)	(1.469)	(1.342)	(1.023)
Establishment Age	0.0367***	0.0309**	0.0375***	0.0095	0.0299	0.0739***	0.0498**	0.0348***
	(0.0080)	(0.0140)	(0.0093)	(0.0178)	(0.0187)	(0.0180)	(0.0243)	(0.0122)
Establishment Age^2	-0.0008***	-0.0010*	-0.0008**	8.7×10^{-5}	-0.0007	-0.0021***	-0.0008	-0.0011**
	(0.0003)	(0.0005)	(0.0003)	(0.0006)	(0.0007)	(0.0006)	(0.0009)	(0.0005)
Old Location Dummy	0.4071***	0.2649**	0.4416***	0.1699	0.3405**	0.6563***	0.7082***	0.2778***
	(0.0632)	(0.1071)	(0.0745)	(0.1447)	(0.1364)	(0.1464)	(0.1845)	(0.0995)
Headquarter Office	0.0342**	0.0521*	0.0277*	-0.0441	0.0079	0.0692*	0.0443*	0.0388
	(0.0140)	(0.0285)	(0.0159)	(0.0308)	(0.0287)	(0.0401)	(0.0254)	(0.0252)
Fixed-effects								
Year	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Pair	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Fit statistics								
F-test (1st stage), Mun. Rate (ln)	256,108.1	87,590.6	196,614.0	45,813.5	50,903.1	37,720.8	63,936.1	80,431.5
Observations	91,282	24,662	66,620	10,371	13,851	19,510	17,918	29,632
Establishments	13,948	4,829	9,119	5,634	1,727	1,993	2,212	2,382
Pairs	29,453	8,126	21,327	10,603	3,364	4,323	5,792	5,371

Clustered (Establishment ID & Second Establishment ID) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Table 5: Estimations of the elasticity of establishment employment with respect to the property tax rate for 2003 - 2011. Column (1) shows the first stage for the combined municipal and educational property tax rate, and column (2) shows the full instrumental variable estimation. Columns (3) and (4) show the same estimation of the municipal and educational tax rate separately as in table 3 Standard errors are clustered by both establishments as an approximation to dyadic robust standard errors. The relevant coefficients are the total rate coefficients and the municipal tax rate coefficient in the top two rows.

Dependent Variables:	Total Rate (ln)	Employment (ln)	Mun. Rate (ln)	Employment (ln)
IV stages	First	Second	First	Second
Model:	(1)	(2)	(3)	(4)
Variables				
Mun. Rate (ln)				0.0004
				(0.2071)
Total Rate (ln)		-0.0044		
		(0.6415)		
Educ. Rate (ln)			2.119***	-0.1663
			(0.0392)	(0.5967)
Hard Cap Policy	0.0458***		0.1414***	
	(0.0030)		(0.0041)	
Establishment Age	-2.11×10^{-5}	0.0367***	-6.16×10^{-5}	0.0367***
	(8×10^{-5})	(0.0080)	(4.37×10^{-5})	(0.0080)
Establishment Age^2	-4.09×10^{-7}	-0.0008***	2.67×10^{-6}	-0.0008***
	(3.06×10^{-6})	(0.0003)	(1.68×10^{-6})	(0.0003)
Old Location Dummy	-0.0006	0.4072***	-0.0003	0.4071***
	(0.0006)	(0.0632)	(0.0004)	(0.0632)
Headquarter Office	6.89×10^{-5}	0.0342**	5.78×10^{-5}	0.0342**
	(0.0001)	(0.0140)	(6.45×10^{-5})	(0.0140)
Fixed-effects				
Year	Yes	Yes	Yes	Yes
Pair	Yes	Yes	Yes	Yes
Fit statistics				
F-test (1st stage)	14,359.2		256,108.1	
F-test (1st stage), Total Rate (ln)		14,359.2		
F-test (1st stage), Mun. Rate (ln)				256,108.1
Observations	91,282	91,282	91,282	91,282
Establishments	13,948	13,948	13,948	13,948
Pairs	29,453	29,453	29,453	29,453

Clustered (Establishment ID & Second Establishment ID) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

One possible concern is that including the education tax rate is introducing endogeneity to the estimation due to the high correlation between the municipal tax rate, which was lowered by the Hard Cap policy, and the education tax rate which was lowered by the provincial government. Removing the education tax rate as a control would lead to an overestimate of the effect of the municipal tax rate, as the same variation in employment would be attributed to a smaller change in property tax rates. Instead this concern can be addressed by combining the municipal and education tax rates into the total tax rate, then applying the same Hard Cap instrument to the combined tax rate.

Table 5 estimates the same within transformed, spatially differenced, and instrumented model using the total tax rate, the sum of the municipal and education tax rates. Column (2) shows the estimate of the elasticity of employment with respect to the total tax rate, and finds it to be indistinguishable from zero, consistent with the estimate for the municipal tax rate alone in column (4). A potential drawback of this approach is that it will reduce the relevance of the instrument, however this is not an issue due to the strong first stage of the instrument.

Given the null estimates thus far, one potential solution would be to expand the distance over which pairing for spatial differencing is occurring. This comes at a tradeoff of increasing the likelihood that spatial differencing will fail to correct for unobserved spatial factors. Evaluating this tradeoff requires some way to measure whether firm outcomes are correlated with some unobserved factor, which is impossible. Instead I make use of observable municipality characteristics as a proxy for unobserved spatial characteristics. If spatial differencing and instrumenting is failing to fully control for spatial factors, then any municipality level control variables should be insignificant. If municipality level control variables are significant then it is likely that the the combination of spatial differencing and instrumenting has failed to control for unobserved spatial factors, and a smaller spatial differencing radius should be used.

To examine whether there are any gains from expanding the distance of spatial differencing, I repeat the fully instrumented specification from column (5) of table 3, but include controls for both population and municipal expenditure at the municipality level. I then repeat the estimation with spatial differencing distances of $1000m$, $2000m$, $3000m$, $4000m$, and $5000m$. Both population and municipal expenditure remain insignificant at the 1000m radius, but become significant for $2000m$ and beyond as seen in tables 7, 8, and 9 of the appendix. This suggests that expanding the matching radius may increase the number of pairs of establishments and therefore the number of observations, but at the cost of increasing bias in the estimations. Thus the $1000m$ radius is retained as the radius for all spatial differencing.

All of the results thus far have also utilized the same time range from 2003-2011, however the Hard Cap policy was most binding in the period between 2001-2004. Panel identifiers in the Scott's database limit estimates to be from 2003 onwards. Table 10 of the appendix repeats the estimations of table 3 using the years 2003-2005, when the Hard Cap policy was most likely to be binding. There are no significant changes to the results. Table 11 of the appendix expands the total time range to 2003-2019 to cover the entire Scott's database, which also finds no significant changes to the results.

In addition to employment I estimate the effect of the property tax on firm export status. Table 6 shows the same specifications as table 3, but with the outcome variable being the export status of establishment. The estimate in table 6 is of the log tax rate on an indicator variable, and the coefficient should be interpreted as the percentage point change in the likelihood of exporting for a one percent change in the tax rate. The coefficient for municipal tax rate on export status is significant, however the magnitude of the estimate is implausible. This can be attributed to spatial heterogeneity not being entirely controlled for, indicated by the coefficient for population in column (5) being significant. The fact that unobserved spatial heterogeneity is not fully controlled for when examining export status could be due to the higher geographic concentration of exporters in small geographic areas, necessitating a radius for spatial differencing of less than 1km. Performing spatial differencing over a smaller radius significantly reduces the number of observations however, making identification of the effect of property taxes on firm export status impossible.

Discussion

Based on the estimates above, the effect of the property tax on firm employment in Ontario is close to zero, with no clear result for the effect on export status of establishments. This implies that non-residential property tax rate changes have little observable effect on establishments. This does not imply that the Ontario property tax reform of 1998 had no impact on establishments, but rather the Hard Cap policies which forced municipalities to adjust their commercial and industrial property tax rates had little effect on establishments. To interpret this result it helps to contrast with the previous literature.

The key mechanism that Duranton et al. (2011) claim results in employment changes for firms as a result of property tax rate changes is the property assessment process. Firms which expand during their sample are forced to revalue their property, resulting in firms declining to expand, in order to avoid paying higher property taxes on their revalued properties. This results in property taxes being not fully capitalized into rental prices. In contrast if property valuations are continuously updated then any change in the property tax rate should be immediately capitalized into non-residential rents, minimizing the effect of any property tax rate change on firm outcomes. A core outcome of the 1998 reform was the continuous updating of property assessments to reflect the market values of property. Through this lens the near zero results for the effect of property tax rates on employment could be viewed as a success. Firms are not being discourage

Table 6: Estimations of the elasticity of establishment export status with respect to the municipal property tax rate for 2003 - 2011. Controls include the education property tax rate, population, municipal expenditure, age of the establishment, age^2 of the establishment, a dummy if the establishment is greater than 20 years old, and whether the establishment is the head office of the firm. Column (1) reports the estimation using pooled OLS, column (2) takes the within transformation to control for establishment fixed effects, and column (3) performs spatial differencing using a 1km threshold and controls for establishment fixed effects. Column (4) reports the first stage, instrumenting for the municipal tax rate using the transformed Hard Cap policy variable, and column (5) reports the full instrumental variable estimation. Standard errors are clustered at the establishment level in columns (1) and (2), and by both establishments in (3), (4), and (5) as an approximation to optimal dyadic robust standard errors. The relevant coefficients are the municipal rate coefficients in the top row.

Dependent Variables:	Export Status			Mun. Rate (ln)	Export Status
IV stages				First	Second
Model:	(1)	(2)	(3)	(4)	(5)
Variables					
Mun. Rate (ln)	-0.0082	-0.0172	-1.355**		-3.010**
	(0.0149)	(0.0153)	(0.6122)		(1.435)
Educ. Rate (ln)	0.1120***	0.0129	1.057	1.150***	2.530
	(0.0318)	(0.0276)	(1.095)	(0.0780)	(1.598)
Population (ln)	0.0431*	-0.0577	1.235**	0.7008***	2.571**
	(0.0240)	(0.0485)	(0.5832)	(0.0617)	(1.281)
Mun. Expenditure (ln)	-0.0439**	-0.0037	-0.2501	0.0337	0.0064
	(0.0222)	(0.0267)	(0.3643)	(0.0448)	(0.3856)
Establishment Age	0.0156***	0.0348***	0.0398***	3.58×10^{-5}	0.0399***
	(0.0056)	(0.0053)	(0.0078)	(2.27×10^{-5})	(0.0078)
Establishment Age^2	-0.0005**	-0.0011***	-0.0013***	-1.04×10^{-6}	-0.0013***
	(0.0002)	(0.0002)	(0.0003)	(8.81×10^{-7})	(0.0003)
Old Location Dummy	0.1623***	0.2864***	0.3114***	0.0003*	0.3122***
	(0.0339)	(0.0393)	(0.0599)	(0.0002)	(0.0599)
Headquarter Office	0.0643***	0.0129**	0.0081	4.28×10^{-5}	0.0082
	(0.0093)	(0.0063)	(0.0074)	(4.91×10^{-5})	(0.0074)
Hard Cap Policy				0.0464***	
				(0.0073)	
Fixed-effects					
Year	Yes	Yes	Yes	Yes	Yes
2 Digit NAICS	Yes	Yes			
Establishment Type	Yes	Yes			
Square Footage Quartile	Yes	Yes			
Establishment ID		Yes			
Pair			Yes	Yes	Yes
Fit statistics					
F-test (1st stage)				15,545.3	
F-test (1st stage), Mun. Rate (ln)					15,545.3
Observations	50,556	50,556	91,282	91,282	91,282
Establishments	13,948	13,948	13,948	13,948	13,948
Pairs	-	-	29453	29453	29453

*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

from expanding due to rate increases, possibly due to the assessment reform.

A second contributing factor for the null employment changes due to the property tax could be due to the assessment base. Both Belotti et al. (2021) and Jofre-Monseny and Solé-Ollé (2010) estimate that the effect of business property taxes, which include other forms of capital, on land and buildings is small, but that there are significant effects on equipment and machinery. In Ontario only the value of the building and land is included in the property tax assessment base. As a result this would suggest that any effect, if found, should be very modest. This finding should also provide a guide to policy makers to not include other forms of capital in local taxes. For example a recent ruling in province of Quebéc which made ”moveable assets” taxable when attached to a building should be a cause for concern (Deschamps, 2022).

Conclusion

This paper has utilized the Ontario property tax reform on 1998 to estimate the effect of the property tax of firms. Using a policy change by the provincial government which forced some municipalities to reduce non-residential property taxes, I find that increases in property taxes rates have no estimated impact on employment. The effect is consistent across property classes and sizes. Disentangling the effects of the reform are challenging because of multiple reforms occurring simultaneously, including updated property assessments, local services reform, and municipal amalgamations. Comparing with the related literature the results suggest that the benefit from the Ontario property reform was the implementation of market value assessment, which ensured that property tax rates do not discourage establishments from growing.

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Appendix

0.1 Robustness

0.1.1 Distance

Table 7: Estimations of the elasticity of establishment employment with respect to the municipal property tax rate for 2003 - 2011. Controls include the education property tax rate, age of the establishment, age^2 of the establishment, a dummy if the establishment is greater than 20 years old, and whether the establishment is the head office of the firm. Column (1) reports the estimation using pooled OLS, column (2) takes the within transformation to control for establishment fixed effects, and column (3) performs spatial differencing using a 1km threshold and controls for establishment fixed effects. Column (4) reports the first stage, instrumenting for the municipal tax rate using the transformed Hard Cap policy variable, and column (5) reports the full instrumental variable estimation. Standard errors are clustered at the establishment level in columns (1) and (2), and by both establishments in (3), (4), and (5) as an approximation to optimal dyadic robust standard errors. The relevant coefficients are the municipal rate coefficients in the top row.

Dependent Variables:	Employment (ln)			Mun. Rate (ln)	Employment (ln)
IV stages				First	Second
Model:	(1)	(2)	(3)	(4)	(5)
Variables					
Mun. Rate (ln)	-0.0966**	-0.0023	-0.1153		-0.2568
	(0.0387)	(0.0219)	(0.8280)		(1.482)
Educ. Rate (ln)	0.3644***	-0.0287	-0.3132	1.150***	-0.1873
	(0.0883)	(0.0458)	(0.9922)	(0.0780)	(1.455)
Population (ln)	0.0168	0.1797***	0.4679	0.7008***	0.5821
	(0.0628)	(0.0644)	(0.6706)	(0.0617)	(1.184)
Mun. Expenditure (ln)	-0.0367	-0.0635	-0.2340	0.0337	-0.2121
	(0.0578)	(0.0389)	(0.4100)	(0.0448)	(0.4722)
Establishment Age	0.0265*	0.0234***	0.0367***	3.58×10^{-5}	0.0367***
	(0.0141)	(0.0061)	(0.0080)	(2.27×10^{-5})	(0.0080)
Establishment Age^2	-0.0002	-0.0005**	-0.0008***	-1.04×10^{-6}	-0.0008***
	(0.0005)	(0.0002)	(0.0003)	(8.81×10^{-7})	(0.0003)
Old Location Dummy	0.6558***	0.2996***	0.4073***	0.0003*	0.4074***
	(0.0873)	(0.0472)	(0.0632)	(0.0002)	(0.0632)
Headquarter Office	0.6852***	0.0182*	0.0342**	4.28×10^{-5}	0.0342**
	(0.0286)	(0.0105)	(0.0140)	(4.91×10^{-5})	(0.0140)
Hard Cap Policy				0.0464***	
				(0.0073)	
Fixed-effects					
Year	Yes	Yes	Yes	Yes	Yes
2 Digit NAICS	Yes	Yes			
Establishment Type	Yes	Yes			
Square Footage Quartile	Yes	Yes			
Establishment ID		Yes			
Pair			Yes	Yes	Yes
Fit statistics					
F-test (1st stage)				15,545.3	
F-test (1st stage), Mun. Rate (ln)					15,545.3
Observations	50,556	50,556	91,282	91,282	91,282
Establishments	13,948	13,948	13,948	13,948	13,948
Pairs	-	-	29453	29453	29453

*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Table 8: Estimations of the elasticity of establishment employment with respect to the municipal property tax rate for 2003 - 2011. Controls include the education property tax rate, age of the establishment, age^2 of the establishment, a dummy if the establishment is greater than 20 years old, and whether the establishment is the head office of the firm. Column (1) reports the estimation using pooled OLS, column (2) takes the within transformation to control for establishment fixed effects, and column (3) performs spatial differencing using a 2km threshold and controls for establishment fixed effects. Column (4) reports the first stage, instrumenting for the municipal tax rate using the transformed Hard Cap policy variable, and column (5) reports the full instrumental variable estimation. Standard errors are clustered at the establishment level in columns (1) and (2), and by both establishments in (3), (4), and (5) as an approximation to optimal dyadic robust standard errors. The relevant coefficients are the municipal rate coefficients in the top row.

Dependent Variables:	Employment (ln)			Mun. Rate (ln)	Employment (ln)
IV stages				First	Second
Model:	(1)	(2)	(3)	(4)	(5)
Variables					
Mun. Rate (ln)	-0.1266***	0.0040	-0.1018		-0.4934
	(0.0351)	(0.0190)	(0.2843)		(0.4005)
Educ. Rate (ln)	0.3433***	-0.0110	-0.5688	1.621***	-0.0783
	(0.0788)	(0.0358)	(0.6781)	(0.0725)	(0.6859)
Population (ln)	0.0047	0.1088*	0.9782***	0.4243***	1.177***
	(0.0580)	(0.0560)	(0.3416)	(0.0473)	(0.4019)
Mun. Expenditure (ln)	-0.0242	-0.0226	-0.4438*	0.0498**	-0.3314
	(0.0533)	(0.0337)	(0.2558)	(0.0240)	(0.2708)
Establishment Age	0.0349***	0.0261***	0.0323***	0.0001	0.0324***
	(0.0123)	(0.0052)	(0.0064)	(0.0001)	(0.0064)
Establishment Age^2	-0.0005	-0.0006***	-0.0007***	-4.98×10^{-6}	-0.0007***
	(0.0005)	(0.0002)	(0.0002)	(3.79×10^{-6})	(0.0002)
Old Location Dummy	0.7158***	0.3013***	0.3843***	0.0005	0.3845***
	(0.0758)	(0.0407)	(0.0516)	(0.0008)	(0.0516)
Headquarter Office	0.6654***	0.0149	0.0250**	0.0002	0.0250**
	(0.0266)	(0.0094)	(0.0110)	(0.0001)	(0.0110)
Hard Cap Policy				0.0800***	
				(0.0060)	
Fixed-effects					
Year	Yes	Yes	Yes	Yes	Yes
2 Digit NAICS	Yes	Yes			
Establishment Type	Yes	Yes			
Square Footage Quartile	Yes	Yes			
Establishment ID		Yes			
Pair			Yes	Yes	Yes
Fit statistics					
F-test (1st stage)				84,230.9	
F-test (1st stage), Mun. Rate (ln)					84,230.9
Observations	62,938	62,938	211,797	211,797	211,797
Establishments	16,880	16,880	16,880	16,880	16,880
Pairs	-	-	66927	66927	66927

*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Table 9: Estimations of the elasticity of establishment employment with respect to the municipal property tax rate for 2003 - 2011. Controls include the education property tax rate, age of the establishment, age^2 of the establishment, a dummy if the establishment is greater than 20 years old, and whether the establishment is the head office of the firm. Column (1) reports the estimation using pooled OLS, column (2) takes the within transformation to control for establishment fixed effects, and column (3) performs spatial differencing using a 5km threshold and controls for establishment fixed effects. Column (4) reports the first stage, instrumenting for the municipal tax rate using the transformed Hard Cap policy variable, and column (5) reports the full instrumental variable estimation. Standard errors are clustered at the establishment level in columns (1) and (2), and by both establishments in (3), (4), and (5) as an approximation to optimal dyadic robust standard errors. The relevant coefficients are the municipal rate coefficients in the top row.

Dependent Variables:	Employment (ln)			Mun. Rate (ln)	Employment (ln)
IV stages				First	Second
Model:	(1)	(2)	(3)	(4)	(5)
Variables					
Mun. Rate (ln)	-0.1182***	0.0143	-0.0955		-0.2794
	(0.0325)	(0.0174)	(0.1168)		(0.1700)
Educ. Rate (ln)	0.3050***	-0.0197	-0.2639	1.989***	0.0466
	(0.0724)	(0.0319)	(0.3575)	(0.0376)	(0.4044)
Population (ln)	0.0189	0.1415***	0.5718***	0.1627***	0.6034***
	(0.0537)	(0.0527)	(0.1572)	(0.0177)	(0.1627)
Mun. Expenditure (ln)	-0.0299	-0.0212	-0.2375**	0.0969***	-0.1657
	(0.0494)	(0.0293)	(0.1087)	(0.0152)	(0.1131)
Establishment Age	0.0368***	0.0246***	0.0278***	0.0002	0.0279***
	(0.0111)	(0.0049)	(0.0065)	(0.0002)	(0.0065)
Establishment Age^2	-0.0006	-0.0006***	-0.0006**	-6.72×10^{-6}	-0.0006**
	(0.0004)	(0.0002)	(0.0002)	(5.73×10^{-6})	(0.0002)
Old Location Dummy	0.7337***	0.2725***	0.3464***	0.0023*	0.3475***
	(0.0680)	(0.0383)	(0.0515)	(0.0012)	(0.0515)
Headquarter Office	0.6691***	0.0143*	0.0249***	0.0002	0.0249***
	(0.0249)	(0.0084)	(0.0095)	(0.0002)	(0.0095)
Hard Cap Policy				0.1013***	
				(0.0024)	
Fixed-effects					
Year	Yes	Yes	Yes	Yes	Yes
2 Digit NAICS	Yes	Yes			
Establishment Type	Yes	Yes			
Square Footage Quartile	Yes	Yes			
Establishment ID		Yes			
Pair			Yes	Yes	Yes
Fit statistics					
F-test (1st stage)				395,426.7	
F-test (1st stage), Mun. Rate (ln)					395,426.7
Observations	73,510	73,510	707,861	707,861	707,861
Establishments	19,323	19,323	19,323	19,323	19,323
Pairs	-	-	217894	217894	217894

*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

0.1.2 Years

Table 10: Estimations of the elasticity of establishment employment with respect to the municipal property tax rate for 2003 - 2005. Controls include the education property tax rate, age of the establishment, age^2 of the establishment, a dummy if the establishment is greater than 20 years old, and whether the establishment is the head office of the firm. Column (1) reports the estimation using pooled OLS, column (2) takes the within transformation to control for establishment fixed effects, and column (3) performs spatial differencing using a 1km threshold and controls for establishment fixed effects. Column (4) reports the first stage, instrumenting for the municipal tax rate using the transformed Hard Cap policy variable, and column (5) reports the full instrumental variable estimation. Standard errors are clustered at the establishment level in columns (1) and (2), and by both establishments in (3), (4), and (5) as an approximation to optimal dyadic robust standard errors. The relevant coefficients are the municipal rate coefficients in the top row.

Dependent Variables:	Employment (ln)			Mun. Rate (ln)	Employment (ln)
IV stages				First	Second
Model:	(1)	(2)	(3)	(4)	(5)
Variables					
Mun. Rate (ln)	-0.1189***	-0.0028	0.1418		0.2649
	(0.0393)	(0.0228)	(0.5632)		(0.6189)
Educ. Rate (ln)	0.4742***	-0.0342	5.056	5.905***	6.329
	(0.1127)	(0.0795)	(10.41)	(0.2691)	(10.94)
Establishment Age	-0.0080	0.0351***	0.0470***	6.12×10^{-5}	0.0470***
	(0.0203)	(0.0100)	(0.0128)	(4.01×10^{-5})	(0.0128)
Establishment Age^2	0.0010	-0.0008**	-0.0013***	-1.43×10^{-6}	-0.0013***
	(0.0008)	(0.0004)	(0.0005)	(1.49×10^{-6})	(0.0005)
Old Location Dummy	0.4065***	0.4020***	0.4029***	0.0007**	0.4027***
	(0.1211)	(0.0827)	(0.1058)	(0.0003)	(0.1058)
Headquarter Office	0.4936***	-0.0119	0.0017	-1.55×10^{-5}	0.0017
	(0.0318)	(0.0134)	(0.0180)	(2.84×10^{-5})	(0.0180)
Hard Cap Policy				0.1325***	
				(0.0022)	
Fixed-effects					
Year	Yes	Yes	Yes	Yes	Yes
2 Digit NAICS	Yes	Yes			
Establishment Type	Yes	Yes			
Square Footage Quartile	Yes	Yes			
Establishment ID		Yes			
Pair			Yes	Yes	Yes
Fit statistics					
F-test (1st stage)				678,796.7	
F-test (1st stage), Mun. Rate (ln)					678,796.7
Observations	21,350	21,350	41,696	41,696	41,696
Establishments	10,675	10,675	10,675	10,675	10,675
Pairs	-	-	20848	20848	20848

*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Table 11: Estimations of the elasticity of establishment employment with respect to the municipal property tax rate for 2003 - 2019. Controls include the education property tax rate, age of the establishment, age^2 of the establishment, a dummy if the establishment is greater than 20 years old, and whether the establishment is the head office of the firm. Column (1) reports the estimation using pooled OLS, column (2) takes the within transformation to control for establishment fixed effects, and column (3) performs spatial differencing using a 1km threshold and controls for establishment fixed effects. Column (4) reports the first stage, instrumenting for the municipal tax rate using the transformed Hard Cap policy variable, and column (5) reports the full instrumental variable estimation. Standard errors are clustered at the establishment level in columns (1) and (2), and by both establishments in (3), (4), and (5) as an approximation to optimal dyadic robust standard errors. The relevant coefficients are the municipal rate coefficients in the top row.

Dependent Variables:	Employment (ln)			Mun. Rate (ln)	Employment (ln)
IV stages				First	Second
Model:	(1)	(2)	(3)	(4)	(5)
Variables					
Mun. Rate (ln)	-0.0894***	0.0284	-0.0503		-0.0355
	(0.0303)	(0.0205)	(0.1912)		(0.2147)
Educ. Rate (ln)	0.3502***	-0.0041	-0.0254	2.145***	-0.0578
	(0.0833)	(0.0372)	(0.4999)	(0.0441)	(0.6356)
Establishment Age	0.0317**	0.0319***	0.0320***	5.2×10^{-5}	0.0320***
	(0.0125)	(0.0056)	(0.0071)	(9.66×10^{-5})	(0.0071)
Establishment Age^2	-0.0004	-0.0008***	-0.0009***	-5.58×10^{-7}	-0.0009***
	(0.0005)	(0.0002)	(0.0003)	(3.26×10^{-6})	(0.0003)
Old Location Dummy	0.6928***	0.3467***	0.3178***	0.0007	0.3178***
	(0.0792)	(0.0420)	(0.0568)	(0.0007)	(0.0568)
Headquarter Office	0.7371***	0.0372***	0.0550***	-8.04×10^{-5}	0.0550***
	(0.0271)	(0.0096)	(0.0130)	(0.0001)	(0.0130)
Hard Cap Policy				0.1424***	
				(0.0041)	
Fixed-effects					
Year	Yes	Yes	Yes	Yes	Yes
2 Digit NAICS	Yes	Yes			
Establishment Type	Yes	Yes			
Square Footage Quartile	Yes	Yes			
Establishment ID		Yes			
Pair			Yes	Yes	Yes
Fit statistics					
F-test (1st stage)				90,911.0	
F-test (1st stage), Mun. Rate (ln)					90,911.0
Observations	70,474	70,474	120,100	120,100	120,100
Establishments	16,228	16,228	16,228	16,228	16,228
Pairs	-	-	34549	34549	34549

*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

0.2 Summary Statistics

Table 12: Summary statistics for tax rates and ratios in 2000.

Variable	Mean	SD	Obs
Commercial Tax Rate 2000	3.46	1.13	532
Industrial Tax Rate 2000	4.98	1.99	502
Residential Tax Rate 2000	1.42	0.33	536
Commercial Tax Revenue 2000	5426121.40	50004421.97	532
Industrial Tax Revenue 2000	1575140.99	10644944.18	499
Residential Tax Revenue 2000	13422126.93	69306540.76	536
Commercial Tax Ratio 2000	1.42	0.41	523
Industrial Tax Ratio 2000	2.12	0.98	479
Residential Tax Ratio 2000	1.00	0.00	536

Table 13: Summary statistics for expenditures by lower tier municipalities in 2000. No expenditure from upper tier municipalities is included.

Expenditure (Thousands)	LT	ST
General government Mean	5366.00	7080.00
General government SD	24131.00	43433.00
Protection services Mean	5709.00	11842.00
Protection services SD	15149.00	78744.00
Transportation services Mean	5856.00	14683.00
Transportation services SD	12369.00	107729.00
Environmental services Mean	6267.00	9420.00
Environmental services SD	14209.00	52063.00
Health services Mean	1748.00	3396.00
Health services SD	2813.00	20332.00
Social and family services Mean	7769.00	18950.00
Social and family services SD	13709.00	113979.00
Social Housing Mean	2118.00	5133.00
Social Housing SD	6166.00	29579.00
Recreation and cultural services Mean	3060.00	6424.00
Recreation and cultural services SD	7404.00	42357.00
Planning and zoning Mean	813.00	1468.00
Planning and zoning SD	1577.00	6253.00
Total Expenditure Mean	38748.00	76722.00
Total Expenditure SD	91962.00	486399.00

0.3 Matching

Merging the FIRs and the Statistics Canada population data requires merging between the two datasets. Taking the FIR data as given in a left join (one to many), matching is first done looking for exact matches based on name. For municipalities that remain unmatched, a regex match is completed where the names that start with the same name as in the FIR data are matched. A final fuzzy matching is completed. Match rates for each step and for the total can be found in table 14.

Table 14: Matching rates between the FIRs and the Statcan population data. The first stage looks for exact matches between municipality name and census subdivision name in a given year, the second stage looks for matches where the municipality name is a substring of the census subdivision name (typical for some names also listen in French). The final join is a fuzzy join where the Jaro–Winkler distance is less than or equal to 0.2, which catches slight differences in spaces or dashes. All missing matches are for Towns or Townships.

Match Stage	Match Rate	Total Match Rate
Exact Matches (ex. 2000)	92.4%	92.4%
Regex Matches (ex. 2000)	25.1%	94.3%
Fuzzy Matches (ex. 2000)	86.8%	99.3%

Merging the Scott's database to each municipality utilizes Statistics Canada's Postal Code Conversion File (PCCF). Table 15 shows the number of firms in the Scott's database and the number of exact matches. Some postal codes are split across multiple municipalities, resulting in more exact matches than the original number of observations. The uniquely identified observations are removed, and observations with multiple possible municipalities are deduplicated using the census subdivision name and the city name listed in the Scott's database, and choosing the closest match via the Jaro–Winkler distance. After deduplicating, the data is merged, giving the total value for matched observations

Table 15: Matching results between Scott's data and the PCCF.

Match Stage	Match Rate	Number of Observations
Scotts Data	NA	611803
Exact Matches	99.8%	740454
Uniquely Identified	91.4%	559818
Multiple Possible CSDs	8.6%	51985
Matched Observations	99.8%	611043

After the Scott's database has been merged to the PCCF giving census subdivisions, each census subdivision is matched to the municipality listed in the FIR data. It is important the the largest municipalities are matched, match rates by municipality type are shown in table 17.

Table 16: Match rates between Scott's database and the FIRs. Note the low rate of municipalities matched to firms, this is in part due to municipalities where no firms are present, such as reserves. The figure also is not exclusive, i.e. a municipality matched in an exact match could also be matched as a fuzzy match. The match rate for firms is much higher as firms tend to concentrate in larger municipalities. I am able to match 99.4% of firms to a municipality. 87.6% of municipalities have matched firms, with Townships missing the most firms.

Match Stage	Municipalities Match Rate	Firm Match Rate	Total Firm Match Rate
Exact Matches	81.8%	97.1%	97.1%
Regex Matches	4.0%	45.5%	98.4%
Fuzzy Matches	5.5%	64.7%	99.4%

Table 17

Municipality Type	Match Rate
C	0.98
County	1
M	0.93
ST	1
T	0.96
Tp	0.79
V	0.93